ERDŐS'S CONJECTURE ON $\sum f(n)/n$ HOLDS FOR EVERY NON-SQUARE MODULUS

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ABSTRACT. Let f be periodic modulo q with $f(n) = \pm 1$ for $q \nmid n$ and $f(n) = 0$ for $q \mid n$. Erdős conjectured, in a letter to Livingston, that $\sum_{n\geq 1} f(n)/n \neq 0$ whenever the series converges; by Baker's theorem the sum is then transcendental. The conjecture was known for prime q, for $2\varphi(q) + 1 > q$ (Okada), for $q \equiv 3 \pmod 4$ (Murty–Saradha) and for a set of $q \equiv 1 \pmod 4$ of lower density at least 0.82 (Chatterjee–Murty). In a companion paper we showed that it fails for $q = 105^2$ and asked whether all failures occur at square moduli. We prove that they do: *Erdős's conjecture holds for every modulus q that is not a perfect square*, and for every such q every convergent sum $\sum f(n)/n$ is transcendental.

The proof is short and does not use computation. It rests on a *parity transfer principle* for Okada's linear criterion for $\sum f(n)/n = 0$. Fix a prime $p \mid q$ and keep only the conditions attached to Dirichlet characters of 2-power order whose conductor is divisible by p. We show that every solution of this subsystem is a superposition of a part supported on the bottom layers, of functions independent of the p-coordinate, and of elementary *telescoping* pairs, each of which moves mass from one divisor layer of $\mathbb{Z}/q\mathbb{Z}$ to the next through an Euler operator $\frac{2^t}{r-1}(r - T_r)$ that annihilates parity. As a consequence, for solutions whose values have constant parity on every layer, the conductor-p projection is congruent modulo 2 to the bottom layer, the residues n with $\gcd(n, q) = q/p^j$, while Okada's logarithmic condition at p forces this projection to be even. For an Erdős function the bottom layers contribute $v_p(q)$ odd terms, so $v_p(q)$ must be even for every p.

The same argument gives a general parity constraint for 2-integral solutions, which we show is sharp: it describes exactly which parity patterns on the divisor layers occur. As a consequence the odd-valued version of the problem is completely classified: for odd $q > 1$ there is a q-periodic f with odd integer values off the multiples of q, vanishing period sum and $\sum f(n)/n = 0$ if and only if q is a square. Together with Okada's theorem and the companion paper this shows that 105^2 is the least counterexample (now without computer assistance), that the conjecture fails for $q < 195^2$ exactly when $q \in \{105^2, 165^2\}$, and that every counterexample modulus is an odd square q with $2\varphi(q) + 1 \leq q$.

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1. Introduction

1.1. **The conjecture.** For an integer $q \geq 2$, an *Erdős function modulo* q is a q-periodic $f\colon \mathbb{Z} \to \{-1, 0, 1\}$ with $f(n) = \pm 1$ for $q \nmid n$ and $f(n) = 0$ for $q \mid n$. Put $L(1, f) = \sum_{n \geq 1} f(n)/n$. The series converges if and only if $\sum_{n=1}^{q} f(n) = 0$; this is a sum of $q - 1$ terms ± 1, so convergence forces q to be odd.

Conjecture 1.1 (Erdős; see Livingston [7])**.** *If f is an Erdős function modulo q and $\sum_{n=1}^{q} f(n) = 0$, then $L(1, f) \neq 0$.*

The question belongs to the circle of problems raised by Chowla [5] on the non-vanishing of $L(1, f)$ for periodic f. For algebraic-valued periodic f the number $L(1, f)$ is either 0 or transcendental, by Baker's theorem on linear forms in logarithms [1, 2, 9]; so Conjecture 1.1 says that every convergent Erdős sum is transcendental. Baker, Birch and Wirsing [2] settled Chowla's question for prime moduli, which proves the conjecture for prime q. Okada [11] turned $L(1, f) = 0$ into a system of $\varphi(q) + \omega(q)$ rational linear equations and proved the conjecture when $2\varphi(q) + 1 > q$. Murty and Saradha [10] proved it for $q \equiv 3 \pmod 4$; Chatterjee and Murty [4] for a set of $q \equiv 1 \pmod 4$ of lower density at least 0.82; Pathak [12] showed that it holds with probability one over Erdős functions; Bharadwaj [3] recast Okada's criterion through primitive characters and found a further family of admissible moduli; Murty and Pathak [8] gave a basis of the solution space of Okada's system. The case $q \equiv 1 \pmod 4$ is described as open in [12, 8, 3].

In a companion paper [6] we showed that the conjecture is *false*: there is an Erdős function modulo $105^2 = 11025$, built from Legendre symbols, with vanishing sum, and there are further counterexamples modulo 165^2, 315^2, 495^2 and 735^2; the set of counterexample moduli is closed under multiplication. The counterexamples were found through a 2-adic principle: since $\pm 1 \equiv 1 \pmod 2$, an Erdős function with $L(1, f) = 0$ is a solution of Okada's system all of whose values are odd. Every odd square has such "odd" solutions, and an exact computation showed that no odd non-square $q \leq 13105$ has one. We conjectured [6, Conj. 5.3] that *no odd non-square has an odd solution*; this would confine all counterexamples to square moduli. The present paper proves this conjecture.

1.2. **Results.** Let $\mathbb{Z}_{(2)}$ be the ring of rationals with odd denominator; its elements are *odd* or *even* according to their residue modulo 2.

Theorem A (Theorem 5.3)**.** *Let $q > 1$ be an integer that is not a perfect square, and let $f\colon \mathbb{Z} \to \mathbb{Z}_{(2)}$ be q-periodic with $f(n)$ odd for $q \nmid n$, $f(n) = 0$ for $q \mid n$, and $\sum_{n=1}^{q} f(n) = 0$. Then $\sum_{n\geq 1} f(n)/n \neq 0$. In particular Conjecture 1.1 holds for every non-square q, and for such q every convergent Erdős sum $\sum f(n)/n$ is transcendental.*

For odd squares the conclusion of Theorem A fails, and the odd-valued version of Erdős's problem is completely settled.

Theorem B (Theorem 5.5)**.** *Let $q > 1$. There is a q-periodic $f\colon \mathbb{Z} \to \mathbb{Z}$ with $f(n)$ odd for $q \nmid n$, $f(n) = 0$ for $q \mid n$, $\sum_{n=1}^{q} f(n) = 0$ and $\sum_{n\geq 1} f(n)/n = 0$ if and only if q is an odd perfect square.*

Theorem A is a special case of a general parity constraint on rational solutions of Okada's system. For a divisor $d > 1$ of q, the *layer* of level d is the set of residues n with $\gcd(n, q) = q/d$.

Theorem C (Parity transfer, Theorem 5.1)**.** *Let q be odd and p a prime divisor of q. Let $f\colon \mathbb{Z} \to \mathbb{Z}_{(2)}$ be q-periodic with $f(0) = 0$, $\sum_{n=1}^{q} f(n) = 0$ and $\sum_{n\geq 1} f(n)/n = 0$, and suppose that on every layer of level d with $p \mid d$ the values of f have a constant parity $c_d \in \{0, 1\}$. Then*

$$\sum_{j=1}^{v_p(q)} c_{p^j} \equiv 0 \pmod 2.$$

Conversely, every family (c_d) satisfying these congruences for all $p \mid q$ is the parity pattern of an integer-valued solution (Theorem 5.2), so Theorem C describes exactly which layer parities occur among the rational solutions of Okada's system. The levels p^j are the layers $\gcd(n, q) = q/p^j$, $1 \leq j \leq v_p(q)$: the residues that are divisible by the full power of every prime of q other than p. For an Erdős function all c_d equal 1 and the sum is $v_p(q)$; taking p with $v_p(q)$ odd gives Theorem A. Theorem C also covers functions with prescribed zeros (Corollary 6.1).

Combining Theorem A with Okada's theorem and with [6] gives the following picture.

Corollary D (Corollary 5.6)**.**
(1) *Every modulus for which Conjecture 1.1 fails is an odd square q with $2\varphi(q) + 1 \leq q$.*
(2) *The least such modulus is $105^2 = 11025$; this statement no longer depends on computer verification.*
(3) *For $q < 195^2 = 38025$ the conjecture fails exactly for $q = 105^2$ and $q = 165^2$; 195^2 is the least modulus for which the truth of the conjecture is unknown.*

1.3. **The principle.** Okada's criterion is a family of linear conditions indexed by the primitive Dirichlet characters χ of conductor dividing q, plus one logarithmic condition for each prime $p \mid q$ (Theorem 2.1). The condition for χ involves the layers d divisible by the conductor of χ, with weights given by the Euler factors $1 - \chi(r)/r$ at the primes $r \mid d$ not dividing the conductor. Pushing every layer forward to the maximal 2-power quotient of $(\mathbb{Z}/d\mathbb{Z})^\times$ keeps all parity information, because the fibres have odd size, and only the characters of 2-power order survive (Section 2).

Fix a prime $p \mid q$ and look only at characters whose conductor is divisible by p. Aggregating the layers with the same set of prime divisors, the conditions say that a family of functions Γ_J on small 2-groups has no "fully primitive" Fourier component, where

$$\Gamma_J = \sum_{L \supseteq J} \prod_{r \in L \smallsetminus J} \frac{r - T_r}{r - 1}\, \pi_{L\to J} \Psi_L,$$

Ψ_L is the aggregated layer, $\pi_{L\to J}$ is a pushforward and T_r is translation by the image of the prime r (Section 3). The operators $\frac{r-T_r}{r-1}$ are not 2-integral. The key observation (Lemma 4.3) is that every solution of these conditions decomposes into a part supported on the bottom layer and elementary solutions of two kinds: functions independent of the p-coordinate, and *telescoping*

pairs, in which a function ω on one layer is balanced by $-R_r\omega$ on the layer obtained by removing the prime r, with

$$R_r = \frac{2^{v_2(r-1)}}{r-1}\,(r - T_r) \in \mathbb{Z}_{(2)}[T_r].$$

In a telescoping pair the non-integral operators cancel exactly, and R_r kills parity: $R_r\omega \equiv (1 - T_r)\omega \equiv 0 \pmod 2$ whenever ω has constant parity. Peeling off elementary solutions from the top layer downwards, one finds that, for a solution whose values have constant parity on each layer, the projection $\Gamma_\emptyset$ to the conductor-p characters is congruent modulo 2 to the bottom layer $\Psi_\emptyset$, which is the sum of the layers of levels $p, p^2, \dots, p^{v_p(q)}$. On the other hand, the characters of conductor exactly p force $\Gamma_\emptyset$ to be constant, and Okada's logarithmic condition at p makes this constant even. So the bottom layer is even. For an Erdős function it is a sum of $v_p(q)$ odd functions, whence $v_p(q)$ is even.

The previously known parity arguments are special cases. Murty and Saradha's proof for $q \equiv 3 \pmod 4$ uses a single identity from the odd characters [10, 6]; the residuacity criterion of [6] is the case in which all translations T_r are trivial, so that no telescoping is needed. Computations in [6] had shown that keeping only quadratic characters, only odd or only even characters, or only characters supported on few primes is not enough; the telescoping structure explains why all characters with p in the conductor must be used together.

1.4. **Organization.** Section 2 recalls Okada's criterion and the pushforward to 2-power quotients. Section 3 sets up the conductor-p subsystem. Section 4 contains the decomposition and telescoping lemmas, which are the heart of the paper. Section 5 proves Theorems A, B, C and Corollary D. Section 6 gives further consequences and remarks, including what remains open for square moduli and for the variant with $f(n) = \pm 1$ for all n. Section 7 lists computations that corroborate the results; they are not used in any proof.

2. Okada's criterion and the pushforward to 2-power quotients

Throughout, $q \geq 3$ is odd, $q = \prod_{p\mid q} p^{e_p}$, and $U_d = (\mathbb{Z}/d\mathbb{Z})^\times$. For a divisor $d > 1$ of q the layer of level d consists of the residues $n = (q/d)x$ with $x \in U_d$; the non-zero residues modulo q are the disjoint union of the layers. For a q-periodic f with $f(0) = 0$ put

$$f_d(x) = f\big((q/d)x\big) \quad (x \in U_d), \qquad S_d = \sum_{x\in U_d} f_d(x), \qquad \langle f_d, \chi\rangle = \sum_{x\in U_d} f_d(x)\overline{\chi(x)}$$

for Dirichlet characters χ modulo d, and for a prime $p \mid q$

$$\varepsilon_p(d) = v_p(q/d) - \frac{[\,p \mid d\,]}{p-1}.$$

Note that $\sum_{n=1}^{q} f(n) = \sum_{d\mid q,\, d>1} S_d$.

Theorem 2.1 (Okada [11]; layered form [3], [6, Prop. 2.1, Thm. 2.2])**.** *Let $f\colon \mathbb{Z} \to \mathbb{Q}$ be q-periodic with $f(0) = 0$ and $\sum_{n=1}^{q} f(n) = 0$. Then $\sum_{n\geq 1} f(n)/n = 0$ if and only if*

(A) *for every non-principal primitive Dirichlet character χ^* whose conductor m divides q,*

$$\sum_{d:\ m\mid d\mid q} \frac{d}{\varphi(d)} \prod_{\substack{r\mid d\\ r\nmid m}} \Big(1 - \frac{\chi^*(r)}{r}\Big)\langle f_d, \chi^*\rangle = 0,$$

where r runs over primes and χ^* is regarded as a character modulo d; and

(B) *for every prime $p \mid q$, $\sum_{d\mid q,\, d>1} S_d\,\varepsilon_p(d) = 0$.*

The sufficiency of (A) and (B) follows from the decomposition

$$\sum_{n\ge1}\frac{f(n)}{n}=\frac{1}{q}\sum_{\chi^*}\Big(\sum_d\frac{d}{\varphi(d)}\prod_{r\mid d,\,r\nmid m}\Big(1-\frac{\chi^*(r)}{r}\Big)\langle f_d,\chi^*\rangle\Big)L(1,\chi^*)-\frac{1}{q}\sum_{p\mid q}\Big(\sum_d S_d\,\varepsilon_p(d)\Big)\log p, \tag{2.1}$$

obtained by grouping $\sum f(n)n^{-s}$ according to $\gcd(n,q)$, expanding each f_d in characters, and letting $s\to1^+$ [6, Prop. 2.1]. Necessity is the substance of Okada's theorem. It combines Baker's theorem on linear forms in logarithms (which separates the logarithms $\log p$ and the even characters, whose $L(1,\chi^*)$ are linear forms in logarithms of cyclotomic units) with the rationality of f and a Galois conjugation argument for the odd characters, whose values $L(1,\chi^*)$ are algebraic multiples of π; see [2, 11], [3, Thm. 4] for exactly this layered form, and the sketch in [6, proof of Thm. 2.2]. Only the necessity of (A) for characters of 2-power order and of (B) at one prime will be used below.

2.1. **Pushforward to 2-power quotients.** For an odd prime r put $t_r=v_2(r-1)\ge1$ and $C_r=\mathbb{Z}/2^{t_r}\mathbb{Z}$. Fix a primitive root g_r modulo r, and for an integer x prime to r let $\kappa_r(x)\in C_r$ be the residue modulo 2^{t_r} of the index of x with respect to g_r modulo r. For $d\mid q$ let

$$Q_d=\prod_{r\mid d}C_r,\qquad \kappa_d\colon U_d\to Q_d,\quad x\mapsto\big(\kappa_r(x)\big)_{r\mid d}.$$

Lemma 2.2. (1) *κ_d is a surjective homomorphism whose kernel H_d has odd order $\varphi(d)/|Q_d|$; it identifies Q_d with the maximal 2-power quotient of U_d.*

(2) *$\chi\mapsto\chi\circ\kappa_d$ is a bijection between the characters of Q_d and the Dirichlet characters modulo d of 2-power order. Writing $\chi=\prod_{r\mid d}\chi_r$ with χ_r a character of C_r, the character $\chi\circ\kappa_d$ is induced by a primitive character of conductor $\prod_{r:\ \chi_r\neq1}r$. In particular the primitive characters of 2-power order and conductor m are the $\chi\circ\kappa_m$ with χ a character of Q_m that is non-trivial on every factor C_r, $r\mid m$; we call such characters of Q_m* full.

Proof. By the Chinese remainder theorem $U_d\cong\prod_{r\mid d}U_{r^{v_r(d)}}$, and U_{r^v} is cyclic of order $r^{v-1}(r-1)$, whose 2-part has order 2^{t_r}. The map $U_{r^v}\to C_r$, $x\mapsto\kappa_r(x)$, is the composite of reduction $U_{r^v}\to U_r$ with the surjection $U_r\cong\mathbb{Z}/(r-1)\to\mathbb{Z}/2^{t_r}$; it is surjective with kernel of order $r^{v-1}(r-1)/2^{t_r}$, which is odd. This proves (1), and (2) follows because a character of C_r composed with κ_r depends only on x mod r and is non-trivial, hence primitive modulo r, unless it is trivial. □

For a q-periodic f with $f(0)=0$ define the *pushforward* of the layer f_d,

$$\Phi_d\colon Q_d\to\mathbb{Q},\qquad \Phi_d(y)=\sum_{x\in U_d:\ \kappa_d(x)=y}f_d(x).$$

Lemma 2.3. (1) *For every character χ of Q_d, $\sum_{y\in Q_d}\Phi_d(y)\overline{\chi(y)}=\langle f_d,\chi\circ\kappa_d\rangle$; in particular $\sum_y\Phi_d(y)=S_d$.*

(2) *If f_d takes values in $\mathbb{Z}_{(2)}$ and has constant parity $c\in\{0,1\}$, then Φ_d takes values in $\mathbb{Z}_{(2)}$ and has constant parity c, and S_d is even.*

Proof. (1) is immediate. (2) Each fibre of κ_d has odd cardinality $|H_d|$, and a sum of an odd number of elements of $\mathbb{Z}_{(2)}$ of parity c has parity c. Finally $S_d=\sum_y\Phi_d(y)$ is a sum of $|Q_d|=2^{\sum_{r\mid d}t_r}$ terms of the same parity, an even number of terms because $d>1$. □

Lemma 2.4 (The logarithmic condition at p)**.** *Let $p\mid q$. If $\sum_d S_d=0$ and $\sum_d S_d\varepsilon_p(d)=0$, then*

$$\sum_{d:\ p\mid d}S_d=-(p-1)\sum_{d:\ p\mid d}v_p(d)\,S_d.$$

If moreover every S_d with $p\mid d$ is even, then $\sum_{p\mid d}S_d\in2^{t_p+1}\mathbb{Z}_{(2)}$.

Proof. Multiply $\sum_d S_d \varepsilon_p(d) = 0$ by $p-1$ and write $v_p(q/d) = e_p - v_p(d)$:

$$(p-1)e_p \sum_d S_d - (p-1)\sum_d v_p(d) S_d - \sum_{p \mid d} S_d = 0.$$

The first term vanishes, and $v_p(d) = 0$ unless $p \mid d$. The last assertion follows since $v_2(p-1) = t_p$. □

3. The conductor-p subsystem

Fix a prime $p \mid q$ and let W be the set of the other prime divisors of q. For $L \subseteq W$ put

$$m_L = p \prod_{r \in L} r, \qquad X_L = Q_{m_L} = C_p \times \prod_{r \in L} C_r.$$

We regard X_L as a product indexed by the set $\{p\} \cup L$ of its *coordinates*. All functions on the groups X_L take values in $\mathbb{Q}$; a function is 2-*integral* if its values lie in $\mathbb{Z}_{(2)}$.

3.1. **Operators.** For $J \subseteq L \subseteq W$ the pushforward $\pi_{L \to J}$ along the coordinate projection $X_L \to X_J$ is $(\pi_{L\to J}F)(y) = \sum_{x \mapsto y} F(x)$. For $r \in W \smallsetminus J$ the prime r is prime to m_J, and we let $h_r^J = \kappa_{m_J}(r) = (\kappa_s(r))_{s \in \{p\}\cup J} \in X_J$ be its image. On functions on X_J define

$$(T_r F)(x) = F(x + h_r^J), \qquad E_r = \frac{r - T_r}{r-1}, \qquad R_r = 2^{t_r} E_r = u_r\,(r - T_r), \quad u_r = \frac{2^{t_r}}{r-1} \in \mathbb{Z}_{(2)}^{\times},$$

and $E_K = \prod_{r \in K} E_r$ for $K \subseteq W \smallsetminus J$ ($E_\emptyset = \mathrm{id}$). We record the elementary properties used below; for a function F on X_J and a character χ of X_J write $\widehat{F}(\chi) = \sum_x F(x)\overline{\chi(x)}$.

- (O1) All T_r, E_r, R_r commute with each other and with pushforwards: $\pi_{L\to J} \circ T_r = T_r \circ \pi_{L \to J}$ for $r \notin L$, because h_r^L projects to h_r^J.
- (O2) $\widehat{T_r F}(\chi) = \chi(h_r^J)\widehat{F}(\chi)$ and hence $\widehat{E_r F}(\chi) = \frac{r - \chi(h_r^J)}{r-1}\widehat{F}(\chi)$. In particular E_r preserves the total sum $\sum_x F(x)$, and maps constants to themselves.
- (O3) R_r is a $\mathbb{Z}_{(2)}$-linear combination of translations, so it maps 2-integral functions to 2-integral functions. If F is 2-integral of constant parity, then $R_r F = u_r(rF - T_r F)$ is even, since r is odd and F and $T_r F$ have the same parity everywhere.
- (O4) Translations preserve the property of being independent of a given coordinate (a function on X_J is *independent of the coordinate* i if it is constant on the fibres of the projection forgetting i).

3.2. **Functions without fully primitive part.** For a finite abelian group $X = \prod_{i \in I} C_i$ with $I \neq \emptyset$ (each C_i cyclic of 2-power order > 1), a character of X is *full* if it is non-trivial on every factor. Let $\mathcal{N}(X)$ be the space of functions F on X with $\widehat{F}(\chi) = 0$ for all full χ, and let $\mathcal{I}_i(X)$ be the space of functions independent of the coordinate i. In Lemma 3.1 functions may be complex-valued; for a $\mathbb{Q}$-valued (or 2-integral) $F \in \mathcal{N}(X)$ the summands can be chosen $\mathbb{Q}$-valued (or 2-integral) by Lemma 4.1(1).

Lemma 3.1. $\mathcal{N}(X) = \sum_{i \in I} \mathcal{I}_i(X)$. *If* $|I| = 1$, $\mathcal{N}(X)$ *consists of the constant functions.*

Proof. A function independent of the coordinate i is orthogonal to every character that is non-trivial on C_i, so $\mathcal{I}_i(X) \subseteq \mathcal{N}(X)$. Conversely, if $F \in \mathcal{N}(X)$, the Fourier expansion of F involves only non-full characters, each of which is trivial on some factor C_i and so lies in $\mathcal{I}_i(X)$. □

3.3. **The aggregated layers.** Let f be q-periodic with $f(0)=0$, let Φ_d be the pushforwards of Section 2, and for $L \subseteq W$ put

$$\Psi_L = \sum_{d\mid q:\ \mathrm{rad}(d)=m_L} \Phi_d, \qquad \text{a function on } X_L.$$

(All layers d with $\mathrm{rad}(d)=m_L$ have $Q_d = X_L$; there are $e_p\prod_{r\in L} e_r$ of them.) In particular $\Psi_\emptyset = \Phi_p + \Phi_{p^2} + \cdots + \Phi_{p^{e_p}}$ is the pushforward of the layers of levels $p, p^2, \dots, p^{e_p}$, which we call the *bottom layers*. For a family $\Psi = (\Psi_L)_{L\subseteq W}$ of functions on the groups X_L and $J \subseteq W$ define the function on X_J

$$\Gamma_J(\Psi) = \sum_{L:\ J\subseteq L\subseteq W} E_{L\smallsetminus J}\,\pi_{L\to J}\Psi_L. \tag{3.1}$$

Proposition 3.2. *If f satisfies condition* (A) *of Theorem 2.1 for all primitive characters of 2-power order whose conductor is divisible by p, then*

$$\Gamma_J(\Psi) \in \mathcal{N}(X_J) \qquad \textit{for every } J \subseteq W.$$

In particular $\Gamma_\emptyset(\Psi)$ is a constant function on C_p, equal to $2^{-t_p}\sum_{p\mid d} S_d$.

Proof. By Lemma 2.2(2) the primitive characters of 2-power order with conductor divisible by p are the $\chi\circ\kappa_{m_J}$, $J\subseteq W$, χ a full character of X_J; the conductor is $m = m_J$. In condition (A) for $\chi^* = \chi\circ\kappa_{m_J}$, a divisor d with $m_J \mid d$ has $\mathrm{rad}(d) = m_L$ for a unique $L \supseteq J$, and then

$$\frac{d}{\varphi(d)} = \prod_{s\in\{p\}\cup L}\frac{s}{s-1}, \qquad \prod_{r\mid d,\ r\nmid m_J}\Big(1-\frac{\chi^*(r)}{r}\Big) = \prod_{r\in L\smallsetminus J}\Big(1-\frac{\chi(h_r^J)}{r}\Big),$$

since $\chi^*(r) = \chi(\kappa_{m_J}(r)) = \chi(h_r^J)$. By Lemma 2.3(1), $\langle f_d, \chi^*\rangle = \widehat{\Phi_d}(\chi\circ\mathrm{pr}) = \widehat{\pi_{L\to J}\Phi_d}(\chi)$, where $\mathrm{pr}\colon X_L \to X_J$ is the projection. Summing over the d with $\mathrm{rad}(d)=m_L$ and dividing by $\prod_{s\in\{p\}\cup J}\frac{s}{s-1}$, condition (A) becomes

$$\sum_{L\supseteq J}\prod_{r\in L\smallsetminus J}\frac{r-\chi(h_r^J)}{r-1}\,\widehat{\pi_{L\to J}\Psi_L}(\chi) = 0,$$

which by (O2) says $\widehat{\Gamma_J(\Psi)}(\chi)=0$. As this holds for every full χ, $\Gamma_J(\Psi)\in\mathcal{N}(X_J)$. For $J=\emptyset$, $X_\emptyset = C_p$ has one coordinate, so $\Gamma_\emptyset(\Psi)$ is constant by Lemma 3.1; by (O2) its total sum is $\sum_L\sum_{x\in X_L}\Psi_L(x) = \sum_{p\mid d} S_d$, and $|C_p| = 2^{t_p}$. □

4. Telescoping

In this section p and W are fixed as in Section 3, and $\Psi = (\Psi_L)_{L\subseteq W}$ denotes an arbitrary family of functions Ψ_L on X_L; it need not come from a periodic function.

4.1. **Integral decomposition.**

Lemma 4.1. *Let $X = \prod_{i\in I} C_i$ be as in Lemma 3.1 and let $F\in\mathcal{N}(X)$ be 2-integral.*

1. *There are 2-integral $F_i \in \mathcal{I}_i(X)$, $i\in I$, with $F = \sum_i F_i$.*
2. *If F has constant parity, the F_i in* (1) *can be chosen so that each of them has constant parity.*

Proof. (1) For $i\in I$ let ϵ_i be the operator that replaces the i-th coordinate of the argument by 0; the ϵ_i are commuting idempotents preserving 2-integrality, and ϵ_i is the identity on $\mathcal{I}_i(X)$. Expanding $\prod_{i\in I}\big(\epsilon_i + (1-\epsilon_i)\big) = 1$ gives $F = \sum_{A\subseteq I} F_A$ with $F_A = \prod_{i\in A}(1-\epsilon_i)\prod_{i\notin A}\epsilon_i F$. For $A = I$ we get $F_I = 0$: by Lemma 3.1, F is a sum of functions in the spaces $\mathcal{I}_i(X)$, and $\prod_{i\in I}(1-\epsilon_i)$

kills each of them. For $A \neq I$, F_A is independent of every coordinate $i \notin A$; assigning each such F_A to one $i \notin A$ gives the F_i.

(2) Let F have constant parity c, take $F = \sum_i F_i$ as in (1), and fix $i_0 \in I$. Let D_i be the function with values in $\{0,1\}$ congruent to F_i modulo 2 for $i \neq i_0$, and to $F_{i_0} - c$ for $i = i_0$; then $D_i \in \mathcal{I}_i(X)$ and $\sum_i D_i \equiv F - c \equiv 0 \pmod 2$. Hence $G = \frac{1}{2}\sum_i D_i$ is 2-integral and lies in $\mathcal{N}(X)$, so by (1) $G = \sum_i G_i$ with 2-integral $G_i \in \mathcal{I}_i(X)$. Then $F_i' = F_i - D_i + 2G_i$ satisfy $\sum_i F_i' = F$, $F_i' \in \mathcal{I}_i(X)$, $F_i' \equiv 0$ for $i \neq i_0$ and $F_{i_0}' \equiv c \pmod 2$. □

4.2. **Elementary solutions.** For $i \in L \subseteq W$ let $\mathrm{pr}_i \colon X_L \to X_{L \smallsetminus \{i\}}$ forget the coordinate i.

Lemma 4.2. (1) (ν-solutions) *Let $\emptyset \neq L_0 \subseteq W$ and let ν be a function on X_{L_0} independent of the coordinate p. Let Ψ have $\Psi_{L_0} = \nu$ and $\Psi_L = 0$ for $L \neq L_0$. Then $\Gamma_J(\Psi) \in \mathcal{N}(X_J)$ for all non-empty $J \subseteq W$, and $\Gamma_\emptyset(\Psi)$ is the constant $2^{-t_p}\sum_{x \in X_{L_0}} \nu(x)$.*

(2) (Telescoping pairs) *Let $i \in L_0 \subseteq W$, $K = L_0 \smallsetminus \{i\}$, and let ω be a function on X_K. Let Ψ have*

$$\Psi_{L_0} = \omega \circ \mathrm{pr}_i, \qquad \Psi_K = -R_i\,\omega, \qquad \Psi_L = 0 \ \ (L \neq L_0, K).$$

Then $\Gamma_J(\Psi) \in \mathcal{N}(X_J)$ for all non-empty $J \subseteq W$, and $\Gamma_\emptyset(\Psi) = 0$.

Proof. (1) Only $L = L_0$ contributes to (3.1), so $\Gamma_J(\Psi) = 0$ unless $J \subseteq L_0$, and then $\Gamma_J(\Psi) = E_{L_0 \smallsetminus J}\pi_{L_0 \to J}\nu$. Since p is a coordinate of X_J that is not summed over, $\pi_{L_0 \to J}\nu$ is independent of the coordinate p, and so is $\Gamma_J(\Psi)$ by (O4); it lies in $\mathcal{N}(X_J)$ by Lemma 3.1. For $J = \emptyset$, $\pi_{L_0 \to \emptyset}\nu$ is a constant function on C_p with total sum $\sum_x \nu(x)$, and E_{L_0} fixes constants (O2).

(2) Let $J \subseteq W$. If $J \not\subseteq L_0$, no term of (3.1) survives and $\Gamma_J(\Psi) = 0$. If $i \in J \subseteq L_0$, only $L = L_0$ contributes, $\pi_{L_0 \to J}$ does not sum over the coordinate i, and so $\Gamma_J(\Psi) = E_{L_0 \smallsetminus J}\pi_{L_0 \to J}(\omega \circ \mathrm{pr}_i)$ is independent of the coordinate i by (O4), hence lies in $\mathcal{N}(X_J)$. Finally let $J \subseteq K$, so that both L_0 and K contribute. Since $\omega \circ \mathrm{pr}_i$ does not depend on the coordinate i and $|C_i| = 2^{t_i}$, we have $\pi_{L_0 \to J}(\omega \circ \mathrm{pr}_i) = 2^{t_i}\pi_{K \to J}\omega$, and $E_{L_0 \smallsetminus J} = E_i E_{K \smallsetminus J}$. Using (O1),

$$\Gamma_J(\Psi) = E_i E_{K \smallsetminus J}\, 2^{t_i}\pi_{K \to J}\omega - E_{K \smallsetminus J}\pi_{K \to J} R_i \omega = E_{K \smallsetminus J}\big(R_i \pi_{K \to J}\omega - R_i \pi_{K \to J}\omega\big) = 0.$$

This covers $J = \emptyset$. □

The cancellation in (2) is the point of the construction: the operator $E_i = \frac{i - T_i}{i - 1}$ is not 2-integral, but in a telescoping pair it only occurs multiplied by $|C_i| = 2^{t_i}$, as the 2-integral operator R_i.

4.3. **The telescoping lemma.**

Lemma 4.3 (Telescoping lemma)**.** *Let $\Psi = (\Psi_L)_{L \subseteq W}$ be a family of 2-integral functions such that*

(i) *Ψ_L has constant parity for every non-empty $L \subseteq W$, and*
(ii) *$\Gamma_J(\Psi) \in \mathcal{N}(X_J)$ for every non-empty $J \subseteq W$.*

Then $\Gamma_\emptyset(\Psi)$ is 2-integral and $\Gamma_\emptyset(\Psi) \equiv \Psi_\emptyset \pmod 2$ at every point of C_p.

No condition is imposed on $\Psi_\emptyset$ or on $\Gamma_\emptyset(\Psi)$.

Proof. Let $h(\Psi)$ be the largest $|L|$ with $L \neq \emptyset$ and $\Psi_L \neq 0$, and $h(\Psi) = 0$ if there is none. We argue by induction on $h = h(\Psi)$. If $h = 0$, then $\Gamma_\emptyset(\Psi) = \Psi_\emptyset$.

Let $h \geq 1$ and let L be any set with $|L| = h$. All terms of $\Gamma_L(\Psi)$ except Ψ_L come from sets strictly larger than L, which carry 0; so $\Psi_L = \Gamma_L(\Psi) \in \mathcal{N}(X_L)$ by (ii). By Lemma 4.1, applied on X_L with coordinate set $\{p\} \cup L$,

$$\Psi_L = \nu_L + \sum_{i \in L} \omega_{L,i} \circ \mathrm{pr}_i,$$

where ν_L is independent of the coordinate p, $\omega_{L,i}$ is a function on $X_{L\smallsetminus\{i\}}$, and all of them are 2-integral of constant parity. Let Ψ' be obtained from Ψ by subtracting, for every L with $|L|=h$, the ν-solution attached to (L,ν_L) and the telescoping pairs attached to $(L,i,\omega_{L,i})$, $i\in L$ (Lemma 4.2). Explicitly, $\Psi'_L=0$ for $|L|\geq h$,

$$\Psi'_K=\Psi_K+\sum_{i\in W\smallsetminus K} R_i\,\omega_{K\cup\{i\},i}\qquad\text{for } |K|=h-1,$$

and $\Psi'_L=\Psi_L$ for $|L|<h-1$. By (O3) each $R_i\omega_{K\cup\{i\},i}$ is 2-integral and even, so Ψ' consists of 2-integral functions, $\Psi'_K\equiv\Psi_K \pmod 2$ for all K, and Ψ' satisfies (i). It satisfies (ii) by linearity and Lemma 4.2. Since $h(\Psi')<h$, the induction hypothesis gives that $\Gamma_\emptyset(\Psi')$ is 2-integral and $\Gamma_\emptyset(\Psi')\equiv\Psi'_\emptyset\equiv\Psi_\emptyset \pmod 2$. By Lemma 4.2 again,

$$\Gamma_\emptyset(\Psi)=\Gamma_\emptyset(\Psi')+\sum_{|L|=h}2^{-t_p}\sum_{x\in X_L}\nu_L(x),$$

and $2^{-t_p}\sum_{x\in X_L}\nu_L(x)=\sum_{y\in\prod_{r\in L}C_r}\nu_L(0,y)$ is a sum of $\prod_{r\in L}2^{t_r}$ elements of $\mathbb{Z}_{(2)}$ of the same parity, an even number of them since $L\neq\emptyset$. Hence $\Gamma_\emptyset(\Psi)$ is 2-integral and $\Gamma_\emptyset(\Psi)\equiv\Psi_\emptyset \pmod 2$. □

Remark 4.4. The proof shows more: every family satisfying (ii) is the sum of a family supported on $L=\emptyset$, of ν-solutions and of telescoping pairs, so these elementary solutions span the solution space of the conditions (ii). For instance, for $p=3$ and $W=\{5,7\}$ (the case $\mathrm{rad}(q)=105$) the 30 unknowns $\Psi_L(x)$ are subject to conditions (ii) of rank 7, leaving a space of dimension 23; we checked the spanning statement numerically in several cases (Section 7).

Example 4.5 (Two primes)**.** Let $W=\{r\}$, write $a=t_p$, $b=t_r$, and let y, z be the coordinates of C_p and C_r. Condition (ii) says that $\Psi_{\{r\}}$ has no full Fourier component, i.e. $\Psi_{\{r\}}(y,z)=\nu(z)+\omega(y)$. Then $\pi_{\{r\}\to\emptyset}\Psi_{\{r\}}=\sum_z\nu(z)+2^b\omega$, and

$$\Gamma_\emptyset(\Psi)=\Psi_\emptyset+E_r\Big(\sum_z\nu(z)+2^b\omega\Big)=\Psi_\emptyset+\sum_z\nu(z)+R_r\,\omega.$$

If $\Psi_{\{r\}}$ has constant parity, one may take ν and ω of constant parity (Lemma 4.1), and then $\sum_z\nu(z)$ and $R_r\omega$ are even. For $q=p^er^f$ with e odd and an Erdős function with $\sum_{n=1}^q f(n)=0$ and $\sum f(n)/n=0$, $\Psi_\emptyset=\sum_{j\leq e}\Phi_{p^j}$ is odd, while $\Gamma_\emptyset(\Psi)$ is an even constant by Proposition 3.2 and Lemma 2.4; this is the whole proof in the case of two primes.

5. Proofs of the main results

Theorem 5.1 (Parity transfer; Theorem C)**.** *Let q be odd, $p\mid q$, and let $f\colon\mathbb{Z}\to\mathbb{Z}_{(2)}$ be q-periodic with $f(0)=0$, $\sum_{n=1}^q f(n)=0$ and $\sum_{n\geq1}f(n)/n=0$. Suppose that for every divisor d of q with $p\mid d$ the layer function f_d has constant parity $c_d\in\{0,1\}$. Then $\sum_{j=1}^{e_p}c_{p^j}\equiv0 \pmod 2$.*

Proof. By Lemma 2.3(2), for $p\mid d$ the pushforward Φ_d is 2-integral of constant parity c_d, and S_d is even. Hence every aggregated layer Ψ_L ($L\subseteq W$) is 2-integral of constant parity, and $\Psi_\emptyset=\sum_{j=1}^{e_p}\Phi_{p^j}$ has constant parity $c=\sum_j c_{p^j} \bmod 2$. By Theorem 2.1, f satisfies (A) and (B), and $\sum_d S_d=0$ by hypothesis. By Proposition 3.2, $\Gamma_J(\Psi)\in\mathcal{N}(X_J)$ for all $J\subseteq W$, and $\Gamma_\emptyset(\Psi)$ is the constant $2^{-t_p}\sum_{p\mid d}S_d$, which lies in $2\mathbb{Z}_{(2)}$ by Lemma 2.4. By the telescoping lemma (Lemma 4.3), $\Gamma_\emptyset(\Psi)\equiv\Psi_\emptyset\equiv c \pmod 2$. Therefore c is even. □

The constraint of Theorem 5.1 is the only one: every parity pattern that it allows occurs.

Theorem 5.2 (Sharpness)**.** *Let q be odd and let $c_d \in \{0,1\}$ be given for every divisor $d > 1$ of q. There is an integer-valued q-periodic f with $f(0) = 0$, $\sum_{n=1}^{q} f(n) = 0$, $\sum_{n\ge1} f(n)/n = 0$, and $f \equiv c_d \pmod 2$ on the layer of level d for every d, if and only if*

$$\sum_{j=1}^{v_p(q)} c_{p^j} \equiv 0 \pmod 2 \qquad \text{for every prime } p \mid q.$$

Proof. Necessity is Theorem 5.1. The patterns satisfying the condition form an $\mathbb{F}_2$-vector space spanned by the indicators e_d of the layers d with at least two distinct prime factors and by the sums $e_{p^i} + e_{p^j}$ $(p \mid q,\, 1 \le i < j \le v_p(q))$; and if f, g realize patterns c, c', then $f+g$ realizes $c+c'$. So it suffices to realize the generators. In both constructions χ is the Legendre symbol modulo a prime $p \mid q$, regarded on $U_{d'}$ for every d' with $p \mid d'$, and f is supported on two layers on which it is a multiple of χ. For such f every period sum $S_{d'}$ vanishes and $\langle f_{d'}, \psi\rangle = 0$ for every primitive $\psi \ne \chi$, so by Theorem 2.1 only condition (A) for χ has to be checked; since $\langle \chi, \chi\rangle = \varphi(d')$ on $U_{d'}$, the term of a layer d' on which $f = \beta\chi$ is

$$\beta\, d' \prod_{s \mid d',\, s \ne p} \Big(1 - \frac{\chi(s)}{s}\Big).$$

(a) Let d have at least two distinct prime factors and let $p \ne r$ be two of them; let $v = v_r(d)$ and $d'' = d/r^v$. Put $f = \chi$ on the layer d and $f = \beta\chi$ on the layer d'', with $\beta = -r^{v-1}\big(r - \chi(r)\big)$. The terms of the two layers are $d''\, r^v \big(1 - \frac{\chi(r)}{r}\big) P$ and $\beta d'' P$ with the same $P = \prod_{s \mid d'',\, s \ne p}(1 - \chi(s)/s)$, so they cancel. Since $r - \chi(r)$ is even, f is odd on the layer d and even elsewhere. (b) For $1 \le i < j \le v_p(q)$ put $f = p^{j-i}\chi$ on the layer p^i and $f = -\chi$ on the layer p^j; the two terms are $p^{j-i}p^i$ and $-p^j$. Here f is odd exactly on these two layers. □

In particular, for $c_d = 1$ for all d (the parity pattern of an Erdős function) the condition says that every $v_p(q)$ is even; for odd q, Theorem 5.2 thus contains both directions of Theorem B below.

Theorem 5.3 (Theorem A)**.** *Let $q > 1$ be an integer that is not a perfect square, and let $f\colon \mathbb{Z} \to \mathbb{Z}_{(2)}$ be q-periodic with $f(n)$ odd for $q \nmid n$, $f(n) = 0$ for $q \mid n$, and $\sum_{n=1}^{q} f(n) = 0$. Then $\sum_{n\ge1} f(n)/n \ne 0$. In particular, Conjecture 1.1 holds for every non-square q, and for every non-square q and every Erdős function f modulo q with $\sum_{n=1}^{q} f(n) = 0$ the number $\sum_{n\ge1} f(n)/n$ is transcendental.*

Proof. If q is even, $\sum_{n=1}^{q} f(n) = \sum_{n=1}^{q-1} f(n)$ is a sum of an odd number of odd elements of $\mathbb{Z}_{(2)}$, so it is not 0 and there is nothing to prove. Let q be odd. Since q is not a square, there is a prime p with $e_p = v_p(q)$ odd. If $\sum f(n)/n$ were 0, Theorem 5.1 would apply with $c_d = 1$ for all d and give $e_p \equiv 0 \pmod 2$, a contradiction. For an Erdős function the sum is therefore a non-zero number of the form $\sum_{a=1}^{q-1} f(a) \cdot \big(-\frac{1}{q}\psi(a/q)\big)$, and such a number is transcendental by Baker's theorem [2, 9]. □

For the converse direction in Theorem B we use the composition lemma of [6].

Lemma 5.4 (Composition [6, Lemma 3.5])**.** *Let f be q-periodic and g be N-periodic, rational-valued, with $f(0) = g(0) = 0$ and $\sum_{n=1}^{q} f(n) = \sum_{n=1}^{N} g(n) = 0$. Define the qN-periodic function $F(n) = f(n)$ for $q \nmid n$ and $F(n) = g(n/q)$ for $q \mid n$. Then $F(0) = 0$, $\sum_{n=1}^{qN} F(n) = 0$ and $\sum_{n\ge1} F(n)/n = \sum_{n\ge1} f(n)/n + \frac{1}{q}\sum_{m\ge1} g(m)/m$.*

Proof. Over a period, $\sum_{q\nmid n} F(n) = N\sum_{n=1}^{q} f(n) = 0$ and $\sum_{q\mid n} F(n) = \sum_{m=1}^{N} g(m) = 0$. The series $\sum_{q\nmid n} f(n)/n = \sum_n f(n)/n$ and $\sum_m g(m)/(qm)$ converge, and their sum is $\sum_n F(n)/n$ (compare partial sums up to x). □

Theorem 5.5 (Theorem B)**.** *Let $q > 1$. There is a q-periodic $f\colon \mathbb{Z} \to \mathbb{Z}$ with $f(n)$ odd for $q \nmid n$, $f(n) = 0$ for $q \mid n$, $\sum_{n=1}^{q} f(n) = 0$ and $\sum_{n\geq 1} f(n)/n = 0$ if and only if q is an odd perfect square.*

Proof. If q is not an odd square, no such f exists by Theorem 5.3 (for even squares by the parity argument at the beginning of its proof). Conversely, for an odd prime ℓ with Legendre symbol $\chi_\ell = (\frac{\cdot}{\ell})$ define the ℓ^2-periodic function

$$g_\ell(n) = \begin{cases} \ell\,\chi_\ell(n/\ell), & \ell \parallel n, \\ -\chi_\ell(n), & \ell \nmid n, \\ 0, & \ell^2 \mid n. \end{cases}$$

Its values off the multiples of ℓ^2 are $\pm\ell$ and ± 1, which are odd, its period sum is $\ell \sum_{a=1}^{\ell-1} \chi_\ell(a) - \ell \sum_{a=1}^{\ell-1} \chi_\ell(a) = 0$, and

$$\sum_{n\geq 1} \frac{g_\ell(n)}{n} = \sum_{\ell\nmid a} \frac{\ell\chi_\ell(a)}{\ell a} - \sum_{\ell\nmid n} \frac{\chi_\ell(n)}{n} = L(1,\chi_\ell) - L(1,\chi_\ell) = 0,$$

where the two series on the right converge, so that splitting the partial sums is legitimate. If $q = \ell_1^2 \cdots \ell_k^2$ (primes ℓ_i not necessarily distinct), applying Lemma 5.4 $k-1$ times to $g_{\ell_1}, \ldots, g_{\ell_k}$ gives a q-periodic integer-valued F with $F(0) = 0$, vanishing period sum and $\sum F(n)/n = 0$; its values at $q \nmid n$ are values of some g_{ℓ_i} at non-multiples of ℓ_i^2, hence odd. □

In the language of [6], Theorem 5.3 says that every odd non-square is 2-*adically obstructed*: Okada's solution space contains no 2-integral vector all of whose coordinates are odd. This is [6, Conj. 5.3], which had been verified there for $q \leq 13105$.

Corollary 5.6 (Corollary D)**.** *Let $\mathcal{C}$ be the set of moduli q for which an Erdős function f with $\sum_{n=1}^{q} f(n) = 0$ and $\sum f(n)/n = 0$ exists.*

1. *Every $q \in \mathcal{C}$ is an odd perfect square with $2\varphi(q) + 1 \leq q$.*
2. *$\min \mathcal{C} = 105^2 = 11025$.*
3. *$\mathcal{C} \cap [1, 195^2) = \{105^2, 165^2\}$, and $195^2 = 38025$ is the least modulus for which it is not known whether it belongs to $\mathcal{C}$.*

Proof. (1) Convergence forces q to be odd, Theorem 5.3 forces it to be a square, and Okada's theorem [11] excludes $2\varphi(q)+1 > q$. (2) $105^2 \in \mathcal{C}$ by [6, Thm. 3.1], whose proof reduces to eleven rational identities. If $q = m^2 < 105^2$ is an odd square, then $m < 105 = 3 \cdot 5 \cdot 7$ has at most two distinct prime factors, so $\varphi(q)/q = \varphi(m)/m \geq \frac{2}{3} \cdot \frac{4}{5} > \frac{1}{2}$ and $q \notin \mathcal{C}$ by (1). (3) Similarly, an odd $m < 195$ with at least three distinct prime factors is 105 or 165, and $165^2 \in \mathcal{C}$ by [6, Thm. 3.3] (verified there by exact computation of all of Okada's equations). For $m = 195 = 3 \cdot 5 \cdot 13$ one has $\varphi(m)/m = 96/195 < \frac{1}{2}$, so neither Okada's theorem nor Theorem 5.3 applies; [6, Sect. 6] shows only that a counterexample modulo 195^2 cannot be invariant under the odd-order units. □

Before this work, part (2) rested on the computer-assisted Theorems 5.6 and 5.9 of [6] (the 2-adic obstruction for all odd non-squares $q \leq 13105$, established by 4938 certificates); it is now a consequence of Theorem 5.3, Okada's theorem and one explicit example.

Remark 5.7. In particular Conjecture 1.1 holds for all moduli outside the set of odd squares $q > 1$, of which there are fewer than $\sqrt{x}/2$ up to x; previously it was known for a set of moduli $q \equiv 1 \pmod 4$ of lower density at least 0.82 [4] (together with all $q \equiv 3 \pmod 4$ [10]). On the other hand $\mathcal{C}$ is infinite, since it contains all powers of 105^2 [6, Thm. 3.6].

6. Further consequences and remarks

6.1. **Prescribed zeros.** Theorem 5.1 applies to functions that vanish on several layers.

Corollary 6.1. *Let q be odd and let $\mathcal{Z}$ be a set of divisors of q containing q. Let f be q-periodic with $f(n) = 0$ if $\gcd(n, q) \in \mathcal{Z}$ and $f(n)$ an odd element of $\mathbb{Z}_{(2)}$ otherwise, and suppose $\sum_{n=1}^{q} f(n) = 0$. If for some prime $p \mid q$ the number of $j \in \{1, \dots, v_p(q)\}$ with $q/p^j \notin \mathcal{Z}$ is odd, then $\sum_{n\geq 1} f(n)/n \neq 0$; if f is rational-valued, the sum is transcendental.*

Proof. On the layer of level d the values of f have constant parity $c_d = [\, q/d \notin \mathcal{Z}\,]$; apply Theorem 5.1. □

For example, let q be odd, let $N \mid q$, and let f be q-periodic with $f(n) = 0$ for $N \mid n$ and $f(n) = \pm 1$ otherwise (so f has the zeros of an Erdős function modulo N but only period q). Here $\mathcal{Z} = \{g \mid q : \ N \mid g\}$, and for each p the number of j with $N \nmid q/p^j$ equals $v_p(N)$. Hence *if N is not a square, every such f with vanishing period sum has $\sum f(n)/n \neq 0$*; for $N = q$ this is Theorem 5.3.

6.2. **Remarks on the proof.**

Remark 6.2 (The value at 0). In Theorems 5.1 and 5.3 the condition $f(0) = 0$ can be dropped for odd q. Indeed, Theorem 2.1 and Lemma 2.4 remain valid if the layer $d = 1$ (the residue 0) is included in the sums over d, with $S_1 = f(0)$ and $\varepsilon_p(1) = e_p$, and the layer $d = 1$ enters neither the conductor-p subsystem nor the sum $\sum_d v_p(d) S_d$.

Remark 6.3 (A pointwise statement). The telescoping lemma needs no hypothesis on the bottom layers. If the layers f_d with $p \mid d$ and d not a power of p have constant parity, and the bottom sums S_{p^j} are even, the proof of Theorem 5.1 shows that for every $y \in C_p$ the number $\sum_{j=1}^{e_p} \sum_{x \in U_{p^j},\, \kappa_p(x) = y} f\big((q/p^j)x\big)$ is even. The hypothesis on the layers that are not powers of p cannot be dropped: for $q = 15$ the integer vector

$$(f(1), \dots, f(14)) = (0, 0, -1, -1, 3, 2, -1, 1, -2, -3, 1, 1, 0, 0)$$

satisfies Okada's conditions (so $\sum f(n)/n = 0$), but $f(5) = 3$ and $f(10) = -3$ are odd while $v_3(15) = 1$; here f has mixed parity on the layers of levels 5 and 15.

Remark 6.4 (Earlier criteria as special cases). For $q \equiv 3 \pmod 4$ the proof of Murty and Saradha [10] (see [6, Prop. 4.4]) uses one identity coming from the odd characters; such q are non-squares, so Theorem 5.3 contains their theorem. For odd prime powers $q = p^e$ the set W is empty and $\Gamma_\emptyset(\Psi) = \Psi_\emptyset$, so no telescoping is needed. The residuacity criterion [6, Thm. 5.4] (some $p \parallel q$ such that all other prime factors of q are 2^{t_p}-th power residues modulo p) is the case in which every translation T_r acting on C_p is trivial; then E_r acts on functions on C_p as the identity and $\Gamma_\emptyset(\Psi) = \sum_L \pi_{L\to\emptyset} \Psi_L$ is visibly congruent to $\Psi_\emptyset$. In general the translations T_r are non-trivial and the operators E_r have denominators; the telescoping lemma shows that these denominators always cancel modulo the conditions attached to the larger conductors.

Remark 6.5 (Which conditions are used). The proof of Theorem 5.3 uses, for one prime p with $v_p(q)$ odd, condition (A) for the primitive characters of 2-power order whose conductor is divisible by p, condition (B) at p, and convergence. The computations reported in [6, Sect. 6] show that this cannot be reduced much further: keeping only quadratic characters, or only characters supported on at most two primes, does not suffice in general.

6.3. **Square moduli.** For a square q every e_p is even, the bottom layers $\Psi_\emptyset$ of an Erdős function are even, and the argument gives no information; indeed odd solutions of Okada's system exist for all odd squares (Theorem 5.5). Whether an Erdős function, with values exactly ±1, exists modulo a square q with $2\varphi(q)+1 \le q$ is a different question, which is not decided by parity. By Corollary 5.6 and [6]:

Problem 6.6. Determine $\mathcal{C}$. It is a multiplicative semigroup of odd squares containing 105^2, 165^2, 315^2, 495^2, 735^2 [6, Thms. 3.1, 3.3, 3.4, 3.6] and contained in $\{q = m^2 : \ m \text{ odd}, \ 2\varphi(q)+1 \le q\}$.

Table 1 lists the nineteen odd $m \le 2000$ with $2\varphi(m^2)+1 \le m^2$; for five of them $m^2 \in \mathcal{C}$ is known, and the other fourteen are open.

TABLE 1. Odd $m \le 2000$ with $2\varphi(m^2)+1 \le m^2$ (Okada's theorem does not apply to $q = m^2$).

status of m^2	m
in $\mathcal{C}$ [6]	105, 165, 315, 495, 735
open	195, 525, 585, 825, 945, 975, 1155, 1365, 1485, 1575, 1755, 1785, 1815, 1995

6.4. **The variant with $f(n) = \pm1$ for all n.** Murty and Pathak [8] considered N-periodic f with $f(n) = \pm1$ for *all* n, including $N \mid n$; convergence then forces N to be even. They proved $L(1,f) \ne 0$ for $N \equiv 2 \pmod 4$, reported Tengely's example of such an f with $L(1,f) = 0$ for $N = 36$ (the least possible period), and left $N \equiv 0 \pmod 4$ open. Our method does not apply directly at odd primes: for even N the layers d with $v_2(d) = 1$ have trivial 2-part $(\mathbb{Z}/2\mathbb{Z})^\times$, and the Euler operator attached to the prime 2 is $\frac{2}{1}\big(1-\frac{T_2}{2}\big) = 2-T_2$, which does not annihilate parity. For instance, for $N = 2p$ the conductor-p conditions say that $\Phi_p + (2-T_2)\Phi_{2p}$ is constant on C_p, and $\Phi_p + (2-T_2)\Phi_{2p} \equiv \Phi_p + T_2\Phi_{2p} \equiv 0 \pmod 2$ holds automatically when $f = \pm1$; this is why Murty and Pathak work at the prime 2 instead. Since $36 = 6^2$, it is natural to ask:

Problem 6.7. Let $N \equiv 0 \pmod 4$ be a non-square and let f be N-periodic with $f(n) = \pm1$ for all n and $\sum_{n=1}^{N} f(n) = 0$. Is $\sum_{n\ge1} f(n)/n \ne 0$?

7. Computational corroboration

The proofs above do not use computation. The following checks, made with exact rational arithmetic (FLINT [13] and Python), corroborate them; the code and logs are in the supplementary material.

7.1. **Agreement with the certificates of [6].** In [6] the non-existence of odd solutions was certified, for each of the 4938 odd composite non-squares $q \le 13105$, by an explicit 2-integral combination of Okada's conditions with odd value on the all-ones vector, and odd solutions were exhibited for the 56 odd squares in that range. Theorem 5.3 explains all of these at once.

7.2. **Support aggregation.** For the existence of odd solutions of the conditions (A) for characters of 2-power order together with (B) and convergence, only three data of q matter: $\mathrm{rad}(q)$, the set O of primes with e_p odd, and the set E_1 of primes with $e_p = 1$. Indeed, (A) depends only on the aggregates of the layers with a common radical, whose parities are $\prod_r e_r \bmod 2$; and the even sums S_d can be redistributed among layers with a common radical without changing the aggregates, which turns (B) at p into $\sum_{p\mid d} S_d = 0$ if $e_p = 1$ and into $\sum_{p\mid d} S_d \equiv 0 \pmod{2^{t_p+1}}$ if $e_p \ge 2$. The resulting systems have $\prod_{r\mid q}(1+2^{t_r}) - 1$ unknowns, independently of the exponents, so one computation covers infinitely many moduli. We decided them by 2-saturation of an exact kernel basis, as in [6, Lemma 4.3].

- For every odd squarefree $r \le 1500$ with at least two prime factors and at most 1200 unknowns, every $O \subseteq \{p \mid r\}$ with $E_1 = \emptyset$, and also $O = E_1 = \{p \mid r\}$: 2178 systems, obstructed exactly when $O \neq \emptyset$.
- For 205 radicals with four to six prime factors from $\{3, 5, \dots, 43\}$, namely the first 177 with four prime factors (in lexicographic order) having at most 2500 unknowns, all 21 with five prime factors and at most 400 unknowns, and all 7 with six prime factors and at most 1200 unknowns, and for $O = \emptyset$, $O = \{p\}$ for each $p \mid r$, $O = \{p \mid r\}$ (all with $E_1 = \emptyset$) and $O = E_1 = \{p \mid r\}$: 1470 systems, obstructed exactly when $O \neq \emptyset$.
- For all 4994 odd composite $q \le 13105$ the aggregated decision agrees with "q is not a square", and for the 404 odd composite $q \le 1200$ it agrees with the decision on the full system of [6, Thm. 4.5].

The full systems of [6, Thm. 4.5] were also solved for some larger moduli: $3^3 5^2 7^2$, $3^2 5^2 7^2 11$, $3^3 5^3 7^2$, $3 \cdot 5^2 7^2 11^2$, $3^3 5^3 7^3$, $3^4 5^3 7^3$ and $3^3 5^4 7^3$ are obstructed, and $3^4 5^2 7^2 = 315^2$ and $3^2 5^2 7^2 11^2 = 1155^2$ are not.

7.3. **The telescoping lemma.**

- For nine configurations (p, W) with up to four primes we formed random superpositions of a bottom layer, ν-solutions and telescoping pairs with random 2-integral data of constant parity, and checked in exact arithmetic that all conditions (ii) of Lemma 4.3 hold, that $\Gamma_\emptyset$ is 2-integral, and that $\Gamma_\emptyset \equiv \Psi_\emptyset \pmod 2$.
- For eight of these configurations (all except $p = 5$, $W = \{13, 17\}$, whose solution space has dimension 151) we compared ranks: the elementary solutions span the full solution space of the conditions (ii), as Remark 4.4 asserts.
- In 353 random tests (random prime sets with two to four primes, random p, and a random constant parity prescribed for every layer Ψ_L, $L \neq \emptyset$), the conductor-p conditions together with the logarithmic condition at p never had a 2-integral solution with odd bottom layer, and always had one with even bottom layer; without the logarithmic condition at p they always had one. So the parity transfer uses exactly the conditions in Theorem 5.1.

7.4. **Sharpness.** For every odd composite $q < 400$ (122 moduli) we computed, from the full system of [6, Thm. 2.2] in all $q - 1$ unknowns, the $\mathbb{F}_2$-space of layer parity patterns of 2-integral solutions; in every case it coincides with the space described by Theorem 5.2.

7.5. **An independent check.** A separate program, written from the statements in this paper and from Okada's original equations [11] (no shared code), confirmed: the rank $\varphi(q)+\omega(q)$ of Okada's system and the existence of odd 2-integral solutions exactly for the squares among 20 moduli $q \le 375$; Theorems 5.1 and 5.2 on the whole mod-2 solution space of these 20 moduli; Proposition 3.2 and Lemma 2.4 on exact solutions for twelve moduli and all their primes (including a control in which reversing the direction of the translations T_r makes the test fail); and Lemmas 4.2 and 4.3, including the spanning statement, in ten configurations (p, W).

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